Interest Calculator Web Application

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Project Type: Academic Mini Project

Abstract

This project focuses on developing a web-based Interest Calculator application using modern web technologies. The application allows users to manage lending and borrowing accounts, calculate simple interest, and track transactions efficiently through a secure and user-friendly interface.

Objectives

- To calculate simple interest accurately.
- To manage user accounts securely.
- To store transaction data in a database.
- To provide an easy-to-use web interface.

Technology Stack

Frontend: React.js
Backend: Node.js with Express.js
Database: MongoDB
Authentication: JWT (JSON Web Token)

API Documentation

POST /api/auth/register – Registers a new user.
POST /api/auth/login – Authenticates user and returns JWT token.
GET /api/account – Fetches user accounts.
POST /api/account – Adds a new account.
GET /api/transaction/:accountId – Fetches transactions of an account.

User Manual

1. User registers using email and password.
2. User logs in to access the dashboard.
3. Dashboard displays account summary.
4. User can add and view accounts.
5. Interest can be calculated using calculator page.
6. User can view transactions and profile details.

Bug Report Sheet

Bug ID	Description	Status
BR-01	Login fails with empty input	Fixed
BR-02	Incorrect interest input accepted	Fixed
BR-03	Page refresh resets state	Known Issue

Conclusion

The Interest Calculator Web Application successfully fulfills the project objectives. It demonstrates full-stack web development concepts including authentication, database integration, and responsive UI design. The project can be extended further with advanced features such as compound interest calculation and notification services.